



okay, I've got 3 software projects all doing the same problem, and I think I'm missing something. They're all distributed meshes of measuring apparatus, whether radio or microphones, or solar wind or EEGs. The issue is taking these spot measurements and extrapolating to a set of sources: atmospheric bounce, a festival setting, the sun, or activity in the brain respectively. In some of these, I can get a raw feed of some of the known sources and model the error but in some I can't. I'm wondering if I'm missing a trick and there's some missing data format or method like modelling the transport that might be a useful data standard. It occurs to me that I might be after the same maths used in Wifi to ignore the bouncing of the signal off walls etc... what might i be missing

You're absolutely on the right track—the unifying concept you're missing is the **forward model** (often expressed as a **Green's function** or **channel impulse response**) that describes how sources propagate through your medium to your sensors. Once you have that, source reconstruction becomes a standard **inverse problem**: you invert the forward operator to recover sources from measurements. ^{[1] [2] [3] [4]}

The common mathematical structure

All four of your systems share the same abstract form:

$$\mathbf{y}(t) = \int G(\mathbf{r}_{\text{sensor}}, \mathbf{r}_{\text{source}}; t - \tau) \mathbf{s}(\mathbf{r}_{\text{source}}, \tau) d\mathbf{r}_{\text{source}} d\tau + \mathbf{n}(t)$$

or in discrete linear-algebra form:

$$\mathbf{y} = \mathbf{G}\mathbf{s} + \mathbf{n}$$

where:

- $\mathbf{y}$ = your sensor measurements (microphones, radio receivers, magnetometers, EEG electrodes)
- $\mathbf{s}$ = the unknown source distribution you want (festival sound sources, ionospheric scatterers, solar wind structures, neural currents)
- $\mathbf{G}$ = the forward operator encoding propagation physics (the “transport model”)
- $\mathbf{n}$ = noise and unmodelled effects

In EEG/MEG this is called the **lead field matrix**; in acoustics it’s the **propagation matrix** or **steering matrix**; in radio it’s the **channel matrix** built from **Channel State Information (CSI)** or **Channel Impulse Response (CIR)**; in atmospheric/space physics it’s often a **Green’s function** for the wave or transport equation. ^{[5] [6] [7] [1]}

Your intuition about Wi-Fi is spot on: Wi-Fi sensing explicitly models multipath via CSI/CIR so it can either **cancel** it (communications) or **exploit** it (sensing/localisation). ^{[7] [8] [9]}

What you might be missing: a standard “transport” data format

There isn’t a single universal file format across all domains, but there *are* standard ways to represent the forward model that you can adopt as your own internal data standard:

1. Channel Impulse Response (CIR) / Channel State Information (CSI)

- **What it is:** Time-domain (CIR) or frequency-domain (CSI) description of how an impulse at the source appears at each sensor. ^{[10] [11] [12]}
- **Used in:** Wi-Fi, UWB, radar, some acoustic systems.
- **Why useful:** Encodes multipath, attenuation, phase shifts, and delays in a form you can convolve with any candidate source.
- **Data shape:** For each transmitter–receiver pair, you store $h(t)$ (CIR) or $H(f)$ (CSI). In MIMO systems this becomes a 3D array: $[N_{\text{tx}}, N_{\text{rx}}, N_{\text{taps}}]$ or $[N_{\text{tx}}, N_{\text{rx}}, N_{\text{subcarriers}}]$. ^{[13] [7]}

If some of your projects don’t yet have this, you can often **estimate** it:

- Send known probes (chirps, impulses, pilot symbols) and deconvolve to get $h(t)$.
- In passive settings, use cross-correlation of ambient noise between sensors to reconstruct an empirical Green’s function. ^[14]

2. Lead field / forward matrix (EEG/MEG style)

- **What it is:** A matrix $\mathbf{G}$ where each column is the sensor pattern produced by a unit dipole at a specific location/orientation in the source space. ^{[6] [15]}
- **Used in:** EEG/MEG, some magnetics, and can be adapted to acoustics and radio if you discretise the source volume.
- **Why useful:** Turns the physics into a linear map $\mathbf{y} = \mathbf{G}\mathbf{s}$ that you can invert with regularisation.
- **Data shape:** $[N_{\text{sensors}}, N_{\text{source_elements}}]$, often precomputed from a head/body/atmosphere model (BEM/FEM or ray-based). ^{[15] [16]}

Even if you don't have MRI-style anatomy, you can build an approximate forward model from:

- Known geometry (sensor positions, approximate source region).
- Simplified physics (spherical head, layered atmosphere, free-space + reflections).

3. Green's function for your PDE

- **What it is:** The impulse response of your governing differential operator (wave equation, diffusion equation, etc.) with your boundary conditions. ^{[3] [1]}
- **Used in:** Acoustics, electromagnetics, space physics, some atmospheric problems.
- **Why useful:** Once you have $G(\mathbf{r}, \mathbf{r}'; t)$, any source distribution's effect is just a convolution.
- **Data shape:** Often stored as a set of precomputed kernels for pairs of source/receiver locations, or parameterised analytically/numerically.

In practice, you might not store the full continuous Green's function but rather a **discretised operator** tailored to your sensor/source grids.

How Wi-Fi “ignores” multipath—and how you can too

In Wi-Fi:

- The channel is modelled as $y(t) = h(t) * x(t) + n(t)$, where $h(t)$ is the CIR capturing all bounces. ^{[12] [10]}
- Receivers estimate $h(t)$ using known pilots, then either:
 - Equalise (invert) it to recover $x(t)$, or
 - Use its structure (CIR/CSI) to infer geometry and motion. ^{[8] [11] [17]}

Your analogue:

- Treat each “bounce” (atmospheric layer, wall, skull, ionosphere) as part of G .
- If you can measure or simulate G , you can:
 - **Deconvolve** to get cleaner source estimates.
 - **Regularise** the inverse to avoid amplifying noise in poorly observed modes.

Where you *can't* directly measure some sources (e.g., unknown brain activity, unknown festival speakers), you're in **blind source separation** territory, but even there, having a forward model for the *propagation* helps immensely. ^{[18] [19] [20]}

Practical steps to unify your three (or four) projects

1. Explicitly define the forward problem for each domain

- Write down: sensors, candidate source space, and governing physics (wave equation, diffusion, quasistatic Maxwell, etc.). ^{[2] [5] [15]}
- Decide on a discretisation: grid, mesh, or basis functions for sources.

2. Compute or estimate a forward operator $\mathbf{G}$

- Analytical/numerical: ray tracing, BEM/FEM, layered-medium solutions, lead-field calculations. ^[16] ^[10] ^[15]
- Empirical: send probes, use cross-correlations, or fit parametric channel models to data. ^[13] ^[14]

3. Adopt a common internal representation

For example, a simple HDF5/NetCDF schema:

- /forward/G – dense or sparse matrix, or a description of how to compute it.
- /forward/geometry – sensor positions, source grid, medium parameters.
- /forward/physics – equation type, boundary conditions, assumptions.
- /data/y – measurements.
- /data/noise_model – optional covariance or noise description.

This becomes your “transport data standard” across projects.

4. Solve the inverse problem with regularisation

- Linear: $\hat{\mathbf{s}} = \arg \min_{\mathbf{s}} \|\mathbf{y} - \mathbf{G}\mathbf{s}\|^2 + \lambda R(\mathbf{s})$
- Use priors that match your domain: sparsity (few sources), smoothness, anatomical constraints, physical bounds. ^[4] ^[21] ^[22]
- Where parts of $\mathbf{G}$ are uncertain, consider **physics-informed neural networks** or Bayesian approaches that treat the forward model itself as uncertain. ^[22] ^[23] ^[24]

5. Leverage blind methods where you lack source labels

- Use **blind source separation (BSS)** or **independent component analysis (ICA)** on top of your forward-model-based preprocessing. ^[19] ^[20] ^[18]
- Your forward model can act as a strong constraint or initialisation for these methods.

Domain-specific pointers

- **Radio / solar wind / ionosphere:**
Look into **channel modelling** and **ray tracing** for plasma/ionospheric propagation; treat CIR/CSI as your core data object. ^[9] ^[10] ^[13]
- **Microphones / festival:**
Acoustic beamforming and BSS literature uses exactly this forward/steering matrix concept; SRP-PHAT and related methods implicitly assume a propagation model. ^[20] ^[18]
- **EEG / brain:**
You’re already in the inverse-problem world; the “missing trick” is often just being explicit about the lead field and using distributed source models with appropriate priors. ^[21] ^[5] ^[6] ^[15]
- **Atmospheric bounce (radio/acoustic):**
Green’s-function-based reconstruction and correlation-based Green’s function estimation are well established in geophysics and could be adapted. ^[25] ^[14]

If you’d like, I can sketch a concrete HDF5 schema or a small Python example showing how to represent $\mathbf{G}$ and solve $\mathbf{y} = \mathbf{G}\mathbf{s} + \mathbf{n}$ in a way that would work across all your projects.

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3. <https://mathworld.wolfram.com/GreensFunction.html>
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